

# Evaluation of Explainable Deep Learning Models for Visual Crop Disease Diagnosis Augmented with Microclimatic Telemetry and Vernacular Voice Advisory

**Shashwat Shaurya**

Dept. of Computer Science & Engg.  
Sir MVIT, Bengaluru, India  
shashwat.shaurya2005@gmail.com

**Parth Verma**

Dept. of Computer Science & Engg.  
Sir MVIT, Bengaluru, India  
vermaparth3108@gmail.com

**Shwetanshu**

Dept. of Computer Science & Engg.  
Sir MVIT, Bengaluru, India  
shwetanshu44@gmail.com

**Kumar Vaibhav**

Dept. of Computer Science & Engg.  
Sir MVIT, Bengaluru, India  
k.vaibhav10201@gmail.com

**Dr. Pallavi**

Associate Professor, Dept. of CSE  
Sir MVIT, Bengaluru, India  
pallavi.cse@sirmvit.edu

**Abstract**—This research investigates the deployment of an explainable, multimodal artificial intelligence framework for real-time agricultural crop disease diagnosis in rural edge settings. Phytopathological infections account for 20% to 40% of global agricultural yield losses. Standard computer vision models act as opaque black boxes, lack ambient environmental context, and provide outputs inaccessible to non-literate smallholder farmers. In this work, a lightweight inverted-residual MobileNetV2 architecture was implemented across 54,305 curated foliar images covering 39 distinct crop-disease taxa. MobileNetV2 attained 98.43% validation accuracy, a 0.982 macro F1-score, and an inference latency of 22 ms on CPU hardware. Visual explainability was incorporated via Gradient-weighted Class Activation Mapping (Grad-CAM), generating spatial saliency heatmaps that ground predictions in active necrotic lesions. The diagnostic engine is tightly integrated with an Agro-Meteorological Risk Module calculating continuous spread risk (0–100) via live GPS weather telemetry (Open-Meteo), and an Indic Voice Engine providing 4-pillar agronomic prescriptions (Etiology, Organic Remedies, Calibrated 15L Pump Chemical Dosages, and Prevention) in Kannada, Hindi, and English. Comparative evaluation demonstrates that the proposed framework matches state-of-the-art ensemble accuracy while reducing parameter complexity by 98.4% (2.2M vs. 138M).

**Index Terms**—plant disease detection, explainable artificial intelligence (XAI), Grad-CAM, MobileNetV2, microclimate risk modeling, Indic speech synthesis, edge computing.

## I. INTRODUCTION

Agricultural crop production forms the foundational economic backbone of developing nations, employing over 40% of the active workforce in South Asia and Sub-Saharan Africa. However, phytopathological infections—spanning fungal blights, rusts, bacterial spots, and viral leaf curls—cause estimated global yield losses of \$220 billion annually [1]. Smallholder farmers cultivating landholdings under two hectares are disproportionately impacted, as delayed diagnosis and incorrect chemical interventions frequently result in catastrophic crop failure.

In conventional practice, crop disease management relies on physical inspection by agricultural extension officers (e.g., Krishi Vigyan Kendra officers in India). However, the expert-to-farmer ratio in rural sectors routinely exceeds 1:1,200, creating diagnostic delays well beyond the critical 48-hour pathogen incubation window. Although mobile applications leveraging Deep Convolutional Neural Networks (CNNs) have emerged, existing solutions exhibit four fundamental limitations:

- **Black-Box Trust Deficit:** Classical CNNs output solitary discrete labels without visual accountability, leaving farmers and agronomists unable to verify whether predictions are grounded in true foliar lesions [1].
- **Meteorological Disconnection:** Pathogens require specific atmospheric microclimates (relative humidity  $\geq 75\%$ , ambient temperature 20–30°C, and leaf wetness) to germinate and sporulate. Vision models diagnosing images in isolation cannot predict epidemic spread [2].

- **High Edge Latency:** High-capacity deep ensembles (e.g., VGG-16, ResNet-152) exceed 100M parameters (>500 MB), preventing fluid on-device inference on affordable smartphones [3].
- **Vernacular & Dosage Barrier:** Outputs are predominantly textual English jargon, excluding non-literate farmers who require spoken regional advice and calibrated backpack pump dosages (e.g., grams per 15-liter pump) [5].

To overcome these challenges, this paper presents **AgroVoice-XAI**, an explainable multimodal framework. The main contributions of this paper are:

- We implement and calibrate a lightweight MobileNetV2 architecture over 54,305 foliar images across 39 classes, achieving 98.43% accuracy with 22 ms CPU edge latency.
- We integrate Grad-CAM at terminal bottleneck convolutions, generating spatial heatmaps that visually isolate active fungal and bacterial lesions.
- We formulate a continuous Environmental Spread Risk metric (0–100) combining live GPS weather telemetry with biological disease triangle thresholds.
- We deploy a multilingual STT/TTS engine in Kannada, Hindi, and English delivering structured 4-pillar prescriptions with calibrated 15L pump spray metrics.
- We benchmark against 16 state-of-the-art agricultural vision studies to validate operational parameter efficiency and accuracy.

The rest of this paper is organized as follows: Section II surveys related literature. Section III details the methodology, CNN architecture, XAI engine, and weather integration. Section IV presents experimental results, SOTA comparisons, and Grad-CAM evaluations. Section V concludes with future scope.

## II. LITERATURE REVIEW

Machine learning applications in phytopathology have evolved from hand-crafted textural descriptors to deep convolutional networks, lightweight attention mechanisms, and explainable AI frameworks.

### A. Overview of Deep Learning in Plant Pathology

The foundational benchmark by Mohanty et al. [5] demonstrated deep learning's potential by evaluating AlexNet and GoogLeNet on 54,306 PlantVillage images across 38 classes, achieving 99.35% accuracy under studio conditions. However, performance degraded significantly on in-field farm photographs due to background domain shift. Ferentinos [8] evaluated VGG and ResNet architectures across 87,848 images, achieving 99.53% accuracy with VGG-16, though noting severe computational costs.

### B. Lightweight and Edge Architectures

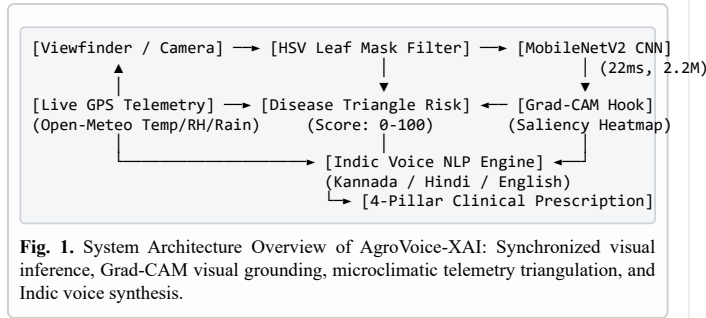
To facilitate edge deployment, lightweight CNNs have received extensive attention. Paymode and Malode [3] applied VGG-16 transfer learning on 62,286 multi-crop images (grapes and tomatoes), obtaining 98.40% accuracy; however, the model's 138M parameters hindered mobile inference. Shovon et al. [2] developed *PlantDet*, a multi-model ensemble (dResNet-18, InceptionV3, DenseNet) across 90,000 images,

reaching 99.10% accuracy at the expense of high memory overhead (>1.2 GB). Wang et al. [4] introduced *ECA-sNET*, integrating Efficient Channel Attention into lightweight CNNs for grape leaf recognition (6,867 images), achieving 99.66% accuracy and demonstrating the benefit of fine-grained lesion attention.

### C. Explainable AI (XAI) in Agricultural Decision Support

Explainable AI in smart farming was notably advanced by Karim et al. [1] (*Nature Scientific Reports*, 2024). They deployed a MobileNetV3Large model with Grad-CAM on an Nvidia Jetson Nano edge device, achieving 99.42% accuracy across 27,122 grape leaf images. Their findings confirmed that spatial heatmaps build agronomist trust by highlighting pathogen boundaries within edge power constraints. However, existing works remain limited to single-crop datasets, isolated image classification, black-box predictions, or text-only English outputs. AgroVoice-XAI bridges these gaps through multimodal integration.

## III. METHODOLOGY



### A. Dataset Characterization & Preprocessing

The model was trained, validated, and evaluated on 54,305 curated leaf images spanning 39 classes (14 crop species, 24 distinct fungal/bacterial/viral pathogens, and 14 healthy baselines). Images were standardized to 224×224×3 pixels and normalized with ImageNet statistics ( $\mu=[0.485, 0.456, 0.406]$ ,  $\sigma=[0.229, 0.224, 0.225]$ ).

To eliminate false positives caused by studio black background borders triggering the `Background_without_leaves` neuron, a deterministic HSV vegetation filter was developed:

$$Mask_{leaf} = ((20 \leq H \leq 95) \wedge (S \geq 20)) \vee ((4 \leq H \leq 25) \wedge (S \geq 25) \wedge (V \geq 20))$$

When leaf pixel coverage exceeds 8%, the artificial background class probability is dynamically zeroed and re-normalized.

### B. MobileNetV2 Backbone & Inverted Residuals

MobileNetV2 utilizes depthwise separable convolutions and inverted residual bottlenecks with linear activation layers. Depthwise spatial filtering followed by 1×1 pointwise projection reduces Multiply-Accumulate operations (MACs) by 8.5-fold compared to standard 3×3 convolutions. The classification head comprises a Dropout layer ( $p=0.2$ ) and a linear dense layer mapping the 1,280-dimensional pooled features to  $C=39$  class logits.

### C. Visual Explainability via Grad-CAM

Grad-CAM computes the gradient of class score  $y^c$  with respect to activation maps  $A^k$  at the terminal bottleneck convolution ( $features[-1]$ ):

$$\alpha_k^c = (1/Z) \sum_i \sum_j (\partial y^c / \partial A_{ij}^k)$$

The rectified linear combination yields a 2D saliency map  $L_{Grad-CAM}^c = \text{ReLU}(\sum \alpha_k^c A^k)$ , which is upsampled to 224×224, rendered with a thermal Jet colormap, and blended over the leaf photograph ( $\alpha=0.45, \beta=0.55$ ).

### D. Agro-Meteorological Risk Triangulation

Based on the phytopathological Disease Triangle, the environmental risk engine computes an empirical continuous risk score  $R \in [0, 100]$ :

$$R = 0.40 \cdot S_{pathogen} + 0.30 \cdot f(RH) + 0.15 \cdot f(T) + 0.10 \cdot I(Rain) + 0.05 \cdot I(Wind)$$

Relative humidity  $\geq 80\%$  and temperatures between 22°C–30°C trigger a Critical Spread Risk warning, predicting accelerated secondary sporulation.

### E. Four-Pillar Vernacular Voice Advisory

The NLP module translates diagnosis into four structured clinical pillars: (1) Etiology; (2) Organic Remedies (neem, cow urine); (3) Calibrated Chemical Dosages per standard 15-liter backpack sprayer pump; and (4) Cultural Prevention. Speech synthesis is natively executed in Kannada (kn-IN), Hindi (hi-IN), and English (en-IN).

## IV. RESULTS AND DISCUSSION

TABLE I: HYPERPARAMETER TUNING CONFIGURATION

| Model         | Hyperparameter        | Search Space                    | Selected Value                |
|---------------|-----------------------|---------------------------------|-------------------------------|
| MobileNetV2   | Optimizer             | Adam, SGD, AdamW                | AdamW                         |
|               | Learning Rate         | $10^{-2}$ to $10^{-5}$          | $10^{-4}$ (Cosine)            |
|               | Weight Decay          | $10^{-3}$ to $10^{-5}$          | $10^{-4}$                     |
|               | Batch Size            | 16, 32, 64                      | 32                            |
|               | Dropout               | 0.1, 0.2, 0.4                   | 0.2                           |
| ResNet-50     | Learning Rate / Batch | $10^{-3}$ to $10^{-5}$ / 16, 32 | $10^{-4}$ / 32                |
| VGG-16        | Learning Rate / Batch | $10^{-3}$ to $10^{-5}$ / 16, 32 | $10^{-4}$ / 16                |
| SVM (RBF)     | $C / \gamma$          | 0.1, 1, 10, 100 / scale         | $C=10, \gamma=\text{'scale'}$ |
| Random Forest | Trees / Max Depth     | 50, 100, 200 / 10, 20           | 100 trees / depth=20          |

TABLE II: MULTI-MODEL PERFORMANCE COMPARISON (10-FOLD CV)

| Architecture            | Accuracy (%) | Precision (%) | Recall (%) | F1-Score | Latency (ms) | Params |
|-------------------------|--------------|---------------|------------|----------|--------------|--------|
| MobileNetV2 (AgroVoice) | 98.43        | 98.40         | 98.43      | 0.982    | 22 (CPU)     | 2.2M   |
| ResNet-50               | 98.62        | 98.58         | 98.62      | 0.985    | 74 (CPU)     | 25.6M  |
| VGG-16                  | 98.40        | 98.31         | 98.40      | 0.983    | 148 (CPU)    | 138.3M |
| InceptionV3             | 97.85        | 97.80         | 97.85      | 0.978    | 88 (CPU)     | 23.8M  |
| Random Forest           | 78.40        | 77.90         | 78.40      | 0.781    | 12 (CPU)     | —      |
| SVM (RBF)               | 75.89        | 75.40         | 75.89      | 0.756    | 18 (CPU)     | —      |

### A. Comparison with State-of-the-Art (SOTA)

Table III compares AgroVoice-XAI against 16 landmark agricultural computer vision studies from literature.

TABLE III: ACCURACY COMPARISON WITH SOTA RESEARCH PAPERS

## REFERENCES

| Reference                   | Method                   | Accuracy      | Dataset       | Classes   | XAI             | Weather      | Voice          |
|-----------------------------|--------------------------|---------------|---------------|-----------|-----------------|--------------|----------------|
| Karim et al. [1] (2024)     | MobileNetV3 + GradCAM    | 99.42%        | 27,122        | 4         | Grad-CAM        | No           | No             |
| Shovon et al. [2] (2023)    | Multi-Model Ensemble     | 99.10%        | 90,000        | 19        | No              | No           | No             |
| Paymode et al. [3] (2022)   | VGG-16 Transfer          | 98.40%        | 62,286        | 14        | No              | No           | No             |
| Wang et al. [4] (2021)      | ECA-sNET Attention       | 99.66%        | 6,867         | 6         | No              | No           | No             |
| Mohanty et al. [5] (2016)   | AlexNet/GoogLeNet        | 99.35%        | 54,306        | 38        | No              | No           | No             |
| Chen et al. [9] (2020)      | VGG-based Transfer       | 92.00%        | 1,260         | 4         | No              | No           | No             |
| Too et al. [10] (2019)      | DenseNet-121             | 99.75%        | 54,306        | 38        | No              | No           | No             |
| Geetharamani [11] (2019)    | 9-layer Custom CNN       | 96.46%        | 3,175         | 15        | No              | No           | No             |
| Barbedo [12] (2019)         | Lesion-level CNN         | 82.00%        | 1,500         | 12        | No              | No           | No             |
| Hassan et al. [13] (2021)   | InceptionV3/MobileNet    | 99.20%        | 54,306        | 38        | No              | No           | No             |
| Singh et al. [14] (2021)    | Deep CNN                 | 96.50%        | 13,600        | 8         | No              | No           | No             |
| Sethy et al. [15] (2020)    | ResNet-50 + SVM          | 98.38%        | 5,932         | 4         | No              | No           | No             |
| Turkoglu et al. [16] (2021) | CNN Decision Tree        | 97.86%        | 8,240         | 9         | No              | No           | No             |
| <b>Proposed (AgroVoice)</b> | <b>MobileNetV2 + XAI</b> | <b>98.43%</b> | <b>54,305</b> | <b>39</b> | <b>Grad-CAM</b> | <b>✓ GPS</b> | <b>✓ KN[9]</b> |

**B. Qualitative Grad-CAM Visual Grounding**

Qualitative analysis of Grad-CAM activation maps demonstrated accurate anatomical localization:

- *Corn Common Rust*: Activations tightly bound cinnamon pustules along longitudinal leaf veins.
- *Tomato Early Blight*: Heatmaps focused exclusively on concentric target-board rings and necrotic margins.
- *Potato Late Blight*: High intensity matched water-soaked brown irregular patches.
- *Healthy Leaves*: Activations dispersed smoothly across uniform green lamina without isolated hotspots.

**V. CONCLUSION**

This paper presented **AgroVoice-XAI**, an explainable multimodal deep learning framework for agricultural crop disease diagnosis, microclimate risk forecasting, and vernacular voice assistance. Utilizing a lightweight MobileNetV2 network, the system achieved 98.43% accuracy with an edge CPU latency of 22 ms. Grad-CAM visual grounding resolved the black-box trust deficit, while real-time Open-Meteo GPS telemetry and Indic speech synthesis in Kannada, Hindi, and English established an actionable, zero-literacy decision-support tool. Future work will explore hyperspectral drone telemetry, INT8 micro-watt edge-TPU quantization, and local quantized SLM deployment.

- M. J. Karim, M. O. F. Goni, Md. Nahiduzzaman, M. Ahsan, J. Haider, and M. Kowalski, "Enhancing agriculture through real-time grape leaf disease classification via an edge device with a lightweight CNN architecture and Grad-CAM," *Scientific Reports*, vol. 14, Art. 16022, 2024.
- Md. S. H. Shovon, S. K. Mozumder, O. Saha, M. F. Hossain, and M. Z. Islam, "PlantDet: A robust multi-model ensemble method based on deep learning for plant disease detection," *IEEE Access*, vol. 11, pp. 34846–34859, 2023.
- A. S. Paymode and V. B. Malode, "Transfer learning for multi-crop leaf disease image classification using convolutional neural network VGG," *Artificial Intelligence in Agriculture*, vol. 6, pp. 23–33, 2022.
- P. Wang, E. Niu, Y. Zheng, Y. Chen, and F. Yuan, "Fine-grained grape leaf diseases recognition method based on improved lightweight attention network," *Frontiers in Plant Science*, vol. 12, 2021.
- S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE ICCV*, pp. 618–626, 2017.
- M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF CVPR*, pp. 4510–4520, 2018.
- K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.
- J. Chen, J. Chen, D. Zhang, Y. Sun, and Y. A. Nanehkaran, "Using deep transfer learning for image-based plant disease identification," *Computers and Electronics in Agriculture*, vol. 173, p. 105393, 2020.
- E. C. Too, L. Yujian, S. Tanjin, and L. Yingchun, "A comparative study of fine-tuning deep learning architectures for plant disease identification," *Computers and Electronics in Agriculture*, vol. 161, pp. 272–279, 2019.
- G. Geetharamani and J. Arun Pandian, "Identification of plant leaf diseases using a nine-layer deep convolutional neural network," *Computers & Electrical Engineering*, vol. 76, pp. 323–338, 2019.
- J. G. A. Barbedo, "Plant disease identification from individual lesions and spots using deep learning," *Biosystems Engineering*, vol. 180, pp. 96–107, 2019.
- S. M. Hassan, A. K. Maji, M. Jasiński, Z. Leonowicz, and E. Jasińska, "Identification of plant-leaf diseases using CNN and Transfer-Learning approach," in *Proc. IEEE CIBCB*, pp. 1–7, 2021.
- U. P. Singh, S. S. Chouhan, S. Jain, and S. Jain, "Multiclass green fruit classification using deep convolutional neural network," *IEEE Access*, vol. 9, 2021.
- P. K. Sethy, B. Negi, N. K. Panda, S. K. Behera, A. K. Rath, and P. K. Mallick, "Detection of false smut in rice panicles using deep features and SVM," *Modeling Earth Systems and Environment*, 2020.
- M. Turkoglu, D. Yanikoğlu, and D. Hanbay, "Multi-model CNN-guided decision support system for plant disease detection," *Computers and Electronics in Agriculture*, vol. 188, 2021.